

ActInf Textbook Group ~ Cohort 3 ~ Meeting 21

(Chapter 9, part 2)

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hi everyone it's cohort three and August 1st we're in our second discussion on chapter nine so if either of you would like before we jump into some specific questions are there any general comments or any other topics that that people want to address otherwise we will uh probably look through some answers and curate and improve some some discourse yep nothing for me all right I I added some questions at the end but we can go through the other stuff first perfect okay and email thank you for your question um what does this text mean about epistemic value so so let me know if it's okay to modify the question to what what is epistemic value if that's what your question was or let me know okay let us uh you added some of the new questions at the at the bottom here Neil yeah that's right all right it's it's one sort of big question really which one Let's Do It um in a non-metronization so that one and then all right awesome yeah all right let's start here all right so so my take on on the chapter is it's it's trying to fit active inference into a a framework of of what science is the the new thing I think the the big idea for me was the metabasian inference which I understand is trying to make uh a so you know someone's trying to make inferences about something has itself got beliefs um so maybe I I don't know if I've got that right so so the first part of that question is is what does what does this have chapter have to contribute when that isn't the case if if we're just trying to use it to understand something that can be modeled and then uh the second part is what the the chapter talks about is is thinking about it in the context of brains uh so Psychiatry but it presumably can be applied to anything in which the the system that's being modeled and in any way be said to be of having beliefs uh and I've I've mentioned then it's intentional stance there where we you know we're not necessarily believing you know we don't think any there's any sentience there so you can think of we could talk about a thermostat thinking it's too hot so uh without actually saying anything about any sort of mental processes going on there um so so it seems to me that the metabase in aspect uh could potentially be saying we could potentially be useful for for for this so beyond Psychiatry psychology yeah and it's so

then skipping over the third question was this question is well is this actually being applied in you know outside of Neuroscience and psychology it seems particularly what I thought be useful to be including it in behavioral economics uh and and maybe in systems engineering I don't know great great questions we'll we'll start with the first ones up ready to go oh oh okay I see the second one was here also welcome Molly thanks and uh I I put the link but this uh upload of the textbook did a little this is a good starting point all right so the first thing that that well let's just start by recalling the The Meta Bayesian approach in nine nine point one so I think you summarized it perfectly Neil were doing Bayesian inference about something that has beliefs so it the aboutness of the subjective model is the environment which may be the laboratory or or the niche um and and thus the aboutness of the scientific model is the rat and the Maze which has an a model that has an aboutness of the maze so that's this kind of like nested or meta Bayesian approach we're using Bayesian approaches from the outside but then when we when we get to the good stuff it is also a Bayesian model with an aboutness of an environment um so I I think this is a great way to put it trying to make inferences about something that has it beliefs or or trying to make inference about something that does inference so your first question was what does the chapter contribute when that is not the case here I believe we can return to our favorite path integrals paper and point to the continuum where on the sophisticated end of the spectrum we have a situation like that in this figure where um you know here's us as scientists here at the blanket States are the solid line and then we're inferring a cognitive model through for example structure learning or or some other um uh a priori constraints we're inferring the cognitive model that's that's looking back onto the so so The metabasian Stance is required in the cognitive on cognitive setting just to be clear that that to model another cognitive entity outside of the metabasing stance I I'm not exactly sure what what the alternative approach would be but you asked what do we get when this is not the case you could model and and and inert um system accordingly here for example maybe here you know it's a light switch or something and it just it just it doesn't have any action selection of its own or it's like inner or it's a passive inference object so what do we gain we gain continuity with those cases thus grounding cognitive Sciences we can discuss this more but but that to me feels like the big home run is continuity along this continuum of of uh sophistication from absolutely inert particles on through arbitrarily complex cognitive entities so it's not to say that it's going to be uniquely explanatory or or of a predictive value it may be like getting on an extra apparatus that you don't necessarily meet which would be the same thing as saying um you could have just started with a simpler model because you're bringing

out this big apparatus that's just condensing down to a simpler model do do any of you think of any other advantages that are gained actually where so where is that that figure from figure two that that paper thank you so it's creating it's creating a scaffolding for the observational encounter that allows us to have system of Interest along this continuity it's absolutely required here you might be able to get away with a simpler perspective in a simpler setting but this is the most expressive way that we can dial it up arbitrarily um complicated and also dial it back down like maybe this just is a single node in this computational graph it just observations get passed in there's a single node and then it gets passed out but this Frame but but the the scientists lab doesn't have to change whether they're studying the metronome um or the bird yeah and I I I I agree that it's important to connect it to these uh like to the standard stages the scientific process Ali or anyone else want to just add more yeah if I may add some other comments about uh but of course I didn't hear the question from the beginning but uh from what I heard uh you see I I believe it's uh essential to um always clarify our Concepts and definitions in terms of what we're really talking about when we talk about uh Agency on one hand um I don't know uh cognition or Consciousness or whatever because all of these terms they have lots of um baggage coming along with them so we need to be careful uh what exactly we mean by agency or in this case I want to emphasize more on agency because this is quite a controversial term because as we know uh in the at least in the mainstream biology people don't want to bestow Agency on uh inert objects or even some simple quote-unquote simple organism but on the other hand there are some mechanical and naturalistic definitions of agency and Welsh's Theory or from fap and so on I I don't know if you've seen this uh nice article by Philip Ball called from a couple of months ago organisms as agents of evolution uh in this pretty much non-technical essay he overviews some of the conceptions about agencies and he also I think uh yeah he he has mentioned fep quite an extensive detail I mean comparatively but the the case here I mean Ball's case is that we still don't know how to model agency as naturalistically as uh I don't know some physical phenomena so of course there are possibilities to model agency these as purely physical trajectory trees or probably to think about and see but I think the point here I want to make is that this is not exactly what we uh always think when we try to trace the path of a physical system through a state space we don't usually think of I don't know simple particle or an inert object as following a path of um or following a goal-directed motion right but on the other hand there was this recent interesting interview Sean Carroll's and Sean Carroll's mindscape podcast with David Krakauer and it was quite quite an interesting podcast because For the First Time I I heard a definition of complex

systems as something that can be directly be applied to fep you know namely he defined complex systems or even the complexity signs as the study of teleonomic matter right so obviously what he meant by teleonomic matter doesn't have anything to do with I don't know material or the black man or pity these come along agents see they HP through [Music] The Rock some weenies this one sorry Ali it just it just it's through the suspected 80. uh sorry just okay can you hear me now yeah just just back up to um chileanomic systems it's not not an extra material cause it's what okay uh so uh yeah it allows for uh the definition of agency through a while to hold spectrum that encompasses uh both in Earth objects or simple organisms up to the most complex phenomenon we know of the uh of in the whole universe namely the human brain uh so this defining characteristic of complex systems as a teleonomic matter I believe uh is a is quite an essential move here because uh this is this can be seen as some of the underlying assertions of fep as well because um the same mathematical technology it I mean if we take Bayesian mechanics as uh something that can be applied both to a simple physical phenomena and to um I don't know Consciousness or a cognition or agency or whatever then we need to uh Define some uh some essential parameter that can be um put in put on to and put onto continuous Spectrum without um on the one hand limiting too much uh and on the other without uh I mean without including or limiting too much so uh yeah these recent moves uh in uh both complexity science and some of the recent advances in biology uh seems interestingly convergent with some of the assertions of fep and I hope in the future uh some interesting unifying perspectives uh can be gleaned from uh seeing these kinds of inter connectedness and interrelation between all of these theories and specifically by looking at how we Define agency in the most naturalistically and mechanistic mechanistic way possible nice great it's like saying like we can't have a framework that has a continuity of ice and liquid and gaseous water it's like well it's not saying that there's not face differences but we want to have something that that like you said doesn't over include or under include and doesn't appeal to extra material or extra systemic factors except credibly and then if we if we um create that measure then we can we can um talk about the agency of different systems okay so I hope but but you know in your question you wrote other than unifying ideas into a common framework so maybe one response is little other than that but that's the big deal um yeah I mean it's it's so it's a big deal for me it's yeah it's it's a beautiful yeah it fits in yeah a nice bigger picture and I yeah I feel as though as you know I've got a lot of a lot of things that could come out of of that framework thinking about it but it's it's not apparent to me what that is at the moment yeah I I mean there's um even just the the concept that some people may

walk around uh feeling and projecting that there's a difference between uh a difference in kinds income measurability difference between like systems that compute or don't or cognitive systems or not agentic systems or not adaptive systems or not sentient systems um it's like well again we can have ice and liquid so it's not like we're we're we're we're neutering our capacity to distinguish systems literally the opposite proponents I'm not even saying anyone specifically is but just just um allegorically proponents of income measurability are actually stymieing their own capacity to compare scientifically they may be able to perform qualitative comparison which is totally legitimate it is what it is though but you cannot have quantitative comparison now it may not be as simple simple as like a scalar number and I think that's kind of the open space with um category Theory and different kinds of equality and sameness but we need to at least have the articulations and and the framework just to scaffold these kinds of systems otherwise we're just begging for there to be a different framework I mean if this was the periodic table could we want a different framework for different particles or I I I'm sure we can think of other ways of where this brings a lot of unity okay next question that you added uh metabasian inference maybe can be applied to any understanding any system of interest in which the subject can be considered having beliefs and intentions then it's intentional stance yeah it including into the the even you have a thermostat um but but but not uh not just a thermostat here as a thermostat coupled to an actor actuator unit because you said it can act but you could say that that yeah yeah a heating system um but this this intentional stance goes even below where most of the intentional stance people would probably feel comfortable saying like that rock just wants to be there hmm yeah and so it's like but that's not intention it's like that's what I'm saying yeah but either you're gonna Grant there to be a Continuum of intentionality and then I think um then whether that's realism is there really a Continuum of intentionality take it or leave it but can we approach systems with a Continuum of absolutely and that at least topologically has to have a structure like this you could have a different um model architecture inside of here um but that model architecture doesn't even have to be what what we would consider necessarily A cognitive model why couldn't this be um uh biological variables and here's the health record and then here's the outcomes of Health so this is even more General than cognitive modeling per se because rather it's just describing the informational relationship between the modeler and the models up to and including systems where the models has an aboutness of the laboratory or of the communication itself um what's the relationship between agency and belief because you could have subjective model that has that could be said to have beliefs but it might not have agency it might just be medical records

for example but those medical records could be said to embody a belief that the the patient has whatever diagnosis yeah that's I I am sure both agency and belief have copious like literature and and semantics just off the off the cuff belief having more to do with what we would associate with the perceptual sense making side of active inference and agency having to do more with the um policy selection aspects like and policy selection entailing uh sentient or adaptive policy selection entailing some belief architecture you could have something that exhibits a lot of agency and has like minimal beliefs like it's just like flailing or babbling or thrashing potentially so it's it's exerting efficacy which under some definitions of agency may qualify with limited belief potentially no Justified true belief whereas in their kind of highest marriage we're talking about like beliefs about action planning as inference those kinds of areas the agency implies action on the environment I think I mean as you say policies it's it's performing some action you know that can then help it learn as opposed to you can have something that just forms beliefs and that's just something doing passive is active model-based analysis actually being applied in areas outside of neuro uh psychop uh psychopathy or psychopathology behavioral economic economic systems engineering so definitely we can bring in some some papers by Henriksen on um cognitive economics now not all cognitive economics is utilizing active but some are this name it sounds familiar and I believe it may be using active inference little but some um industrial systems engineering yes uh well robotics another area industrial systems engineering ranging from the most logistical and operational like what verses and others are involved in uh on through like what um Stephen Fox is involved in with uh social organization so here's this then here we have the Stephen Fox John boyck angle with socio-technical systems Niagara ecology in fact I I I'm gonna I'm gonna go to another um this this is a little little peep into our uh into our our open source ecosystems document um which of course Ali and and Neil you're welcome to be authors on just just uh email me if you if you don't have the information that you feel like you need this is our citation train I think we went from we go from yeah literally citation 7 to 99 so it's kind of a monster cognitive neuroscience and cognitive philosophy artificial intelligence and AI explainability robotics human health clinical Psychiatry social developmental psychology social sciences mathematical physics physics of life Consciousness studies phenomenology evolutionary ecology and development cyber and cognitive security mapping and modeling of information ecosystems ontology modeling and maintenance rhetorical analysis Logistics and business intelligence theoretical and Applied Biology and ecology project management and team science Collective Behavior military science if there are some other domains um we will

literally add them in fact we probably should add economics um I know that we cite some though but also I'll just write social sciences and any thoughts on this um list or or like what other domains would be exciting for you to see that's your education I I'm just gonna I'm just gonna copy it into the chat too yeah that's too long it's too long I mean uh but just uh I'm gonna copy the uh uh the document there but but yeah of course we're the textbook groups I'm not gonna go down the rabbit hole with the open source ecosystems but um that's that's exactly what we've been thinking about and and also we didn't even like cite digital Gaia versus and so on um or thought Forge or any of our other um colleagues in industry and that's kind of the idea with the ecosystem we have people who are working more on the theory and then it's not always so visible necessarily where the applications are happening but I think we can do a lot of work to improve like the discoverability and connect you know use the ontology to connect across settings and cases there's there's just a lot but again another conversation all right um esmail asked what is epistemic value let's see where it comes up in um chapter nine otherwise we can look look to an earlier chapter as well yeah epistemic values primarily discussed in in chapter seven Ali wanna what's your take on on epistemic value and Nails well just let let's just get some uh looking from chapter nine looking back to seven what is epistemic value okay so epistemic value is uh one of the a key terminology of active inference which is also in parallel with um I mean practical value because you see sometimes uh when an agent tries to infer its best course of action within an environment it's not always a trade-off between the thing that or it's it's not a a necessary trade-off between survival and also the minimum the explore exploit phenomena so for example in some of the cases the agent needs to somehow gather information about the environment and it and it actually prefers Gathering the information about the environment and the the situation in which it resides uh in order to take more um suitable actions uh for it to to preserve its persistence through time uh or in other words it's survival uh so epistemic value is what uh gives um a kind of numerical value to this kind of exploring uh the environment and also how we can use this numerical uh value uh in a trade-off between what we gain in uh what what we gain informationally about the environment and the course of actions we take through within that environment so sometimes it acts as a central essential goal or purpose for the agent to increase its epistemic value through its seek information seeking actions rather than just changing the environment in order to better fit its generative model nice thank you Neil what's your take oh you're muted if you're adding anything or all sorry I was yeah yeah I was thinking when I was talking I we think I I think of the this anxiety exploration exploitation um so in reinforcement learning uh

it's it's very much exploration is is for direct impact on the policy um but we as humans go through life collecting information that and storing it with likely um no future use of it so you could say we we're epistemic we do epistemic Gathering of stuff that has no apparent epistemic value but circumstances might change and it might become valuable um so I don't know if there's a distinction to be made between Well yeah if bringing in this idea of of knowledge yeah that that's a very interesting point about like the pure value of information at least with respect to its role in decision making like if we're thinking about people using social media applications and so on or or um or or or who knows what else maybe even um uh just the the desire to to flip every rock over see if there's an ant colony underneath there like there are um uh expected free energy being composed of pragmatic and epistemic value gives us extreme flexibility to Model A diversity of of settings and cases so this is summarized in um figure 2.6 on the left of the main equation we have pragmatic value which is how surprising observations are with respect to preferences and on the right side we have the larger epistemic value which is a KL Divergence hence can be attracted practically optimized in the variational Bayesian and three and five are identical whereas two and four differ and specifically four brings in the observations as a conditioned upon that specific policy being enumerated or calculated for expected free energy so under a policy where the observations are not expected to to add any value in terms of reducing uncertainty that policy is not assigned a high epistemic value it may have a high pragmatic value or not whereas a sequence of observations that a given policy provides which do cause a Divergence from what is what is understood other than through that action being taken that would be informative again that may positive or A A detrimental pragmatic value like what's like I wonder what would happen if I did this but that may threaten homeostasis but having epistemic and pragmatic value Unified under a singular decision-making imperative allows us to find these previously unappreciated Concepts um in in a uh to to be under the umbrella of expected free energy which we utilize for policy selection but most briefly epistemic values the value of incoming information in terms of observations about hidden States reducing uncertainty about the hidden States chapter 7 deals with it more one interesting question maybe for the future is like how is epistemic value treated differently in the discrete and in the continuous case like do you need explicit planning discrete potentially maybe even symbolized planning in order to truly engage with the counterfactual value of information so how do we how do we um think about the epistemic value of continuous activity like groping I mean opt there might be a connection to Optimal grasp and groping around in a sort of continuous reflexive way to to gain a grip on an object

but with that load on pragmatic value where optimal grip is preferred and or without load on epistemic value where how to achieve optimal grasp or what it would feel like to have optimal grasp is not known what are the two related reasons for fitting a computational model to observed Behavior I believe this is an exact quote Let's uh let's see how well perplexity.ai does it's going to have to context switch a little bit it's going to have to change languages but we all do I think this is also which more and more uh synthetic intelligences are gonna now it's asking us us a question now how would it evaluate all the questions that it could ask us which one should ask well active inference how will it know what question to ask what path of action is going to be most relevant okay here we go here we go first estimate parameters of Interest in terms of computational phenotyping the second is to compare alternative hypotheses so like to give an example from from botany the first one would be like we have a bunch of strains of corn and we want to know which one has a different growth rate computational phenotyping the second one would be like does corn growth rate depend on just humidity or humidity and genotype or humidity genotype and temperature so the first one is parameter estimation computational phenotyping is parameter estimation and the second reason the comparison of alternative hypotheses is model comparison the synthetic intelligence here has provided uh a difference set of answers that aren't aren't aren't terrible and it can't necessarily answer perfectly from the textbook anyways 9.1 wow Ollie this looks like somewhat of a little side jury I don't remember it but it seems like it happens foundationalists coherentists infinitists that's good to have about that this goes beyond proof of principle simulations we have seen in previous chapters instead exploits active inference in answering scientific questions what does that mean this is a pretty concise answer previous chapters generated the model and played around with it EG used it to generate plausible data now we are building the Machinery to take in real data and fit that to data and and fit that data to a model to slash with the model two uses on page 174. map territory fallacy not a full question how does this work okay how does the work connect with map territory fallacy in the context of chapter nine being about data analysis does the work commit the map territory fallacy or rather help us articulate or evade being fallacious so let's think about figure 9.1 some have criticized fep by implying or suggesting it's guilty of making a fallacy that the map is the territory I mean citations needed Also regarding generative process and generative model uh we actually spoke with Maxwell or in a scientific Advisory board meeting it was really funny and of course um Illuminating of high epistemic value here's what he added this morning I I of course pointed to Ali's uh question during the uh inner screen stream and he said there's one generative model it's the

system level description it's our model of the whole environment agent system it is the entire causal connective tissue it's a joint density of the real model of the dependencies of the system if the generative model contains a Markov blanket via sparsity and it does then you get an additional density the variational density where internal states parameterize with an aboutness of external States and those internal States will look like they are tracking external things in the sense of the generalized synchrony generative process refers to the Dynamics that give rise to observations that are hard coded in and so he analogized that to the Newton's Laws it is not the case that the generative model is a model of the generative process philosophers are comfortable with approaches like the helmholtz in the Bolton machine and from there arose the distinction between the recognition and the generative density the tail to densities but that is not the sense in which generative model is being used here so just kind of a fun update to an ongoing discussion and Carl was there ends endorsed and approved time flies time flies like an arrow fruit flies like a banana as the famous quote said we are at the end of chapter nine now in the coming weeks we will be heading into chapter 10. then after two weeks on the uh text and concept heavy chapter 10 we will be closing this cohort so it's very exciting presumably after this we will wrap the transcripts of cohort one two and three and publish those three together anyone have any closing thoughts good session all right next next time chapter 10 looks good awesome yep totally looking forward to thank you fellas till next time thanks cheers bye